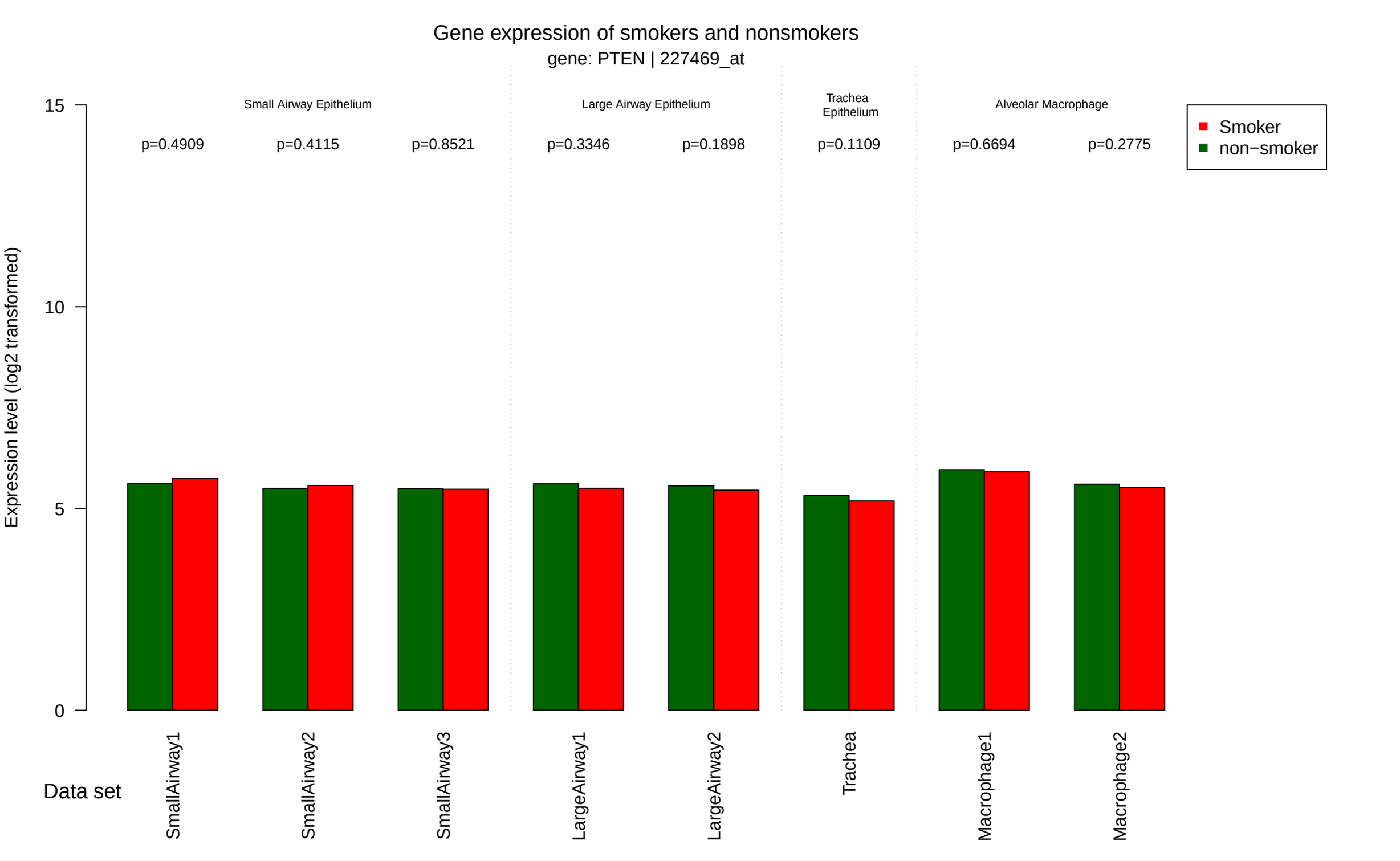
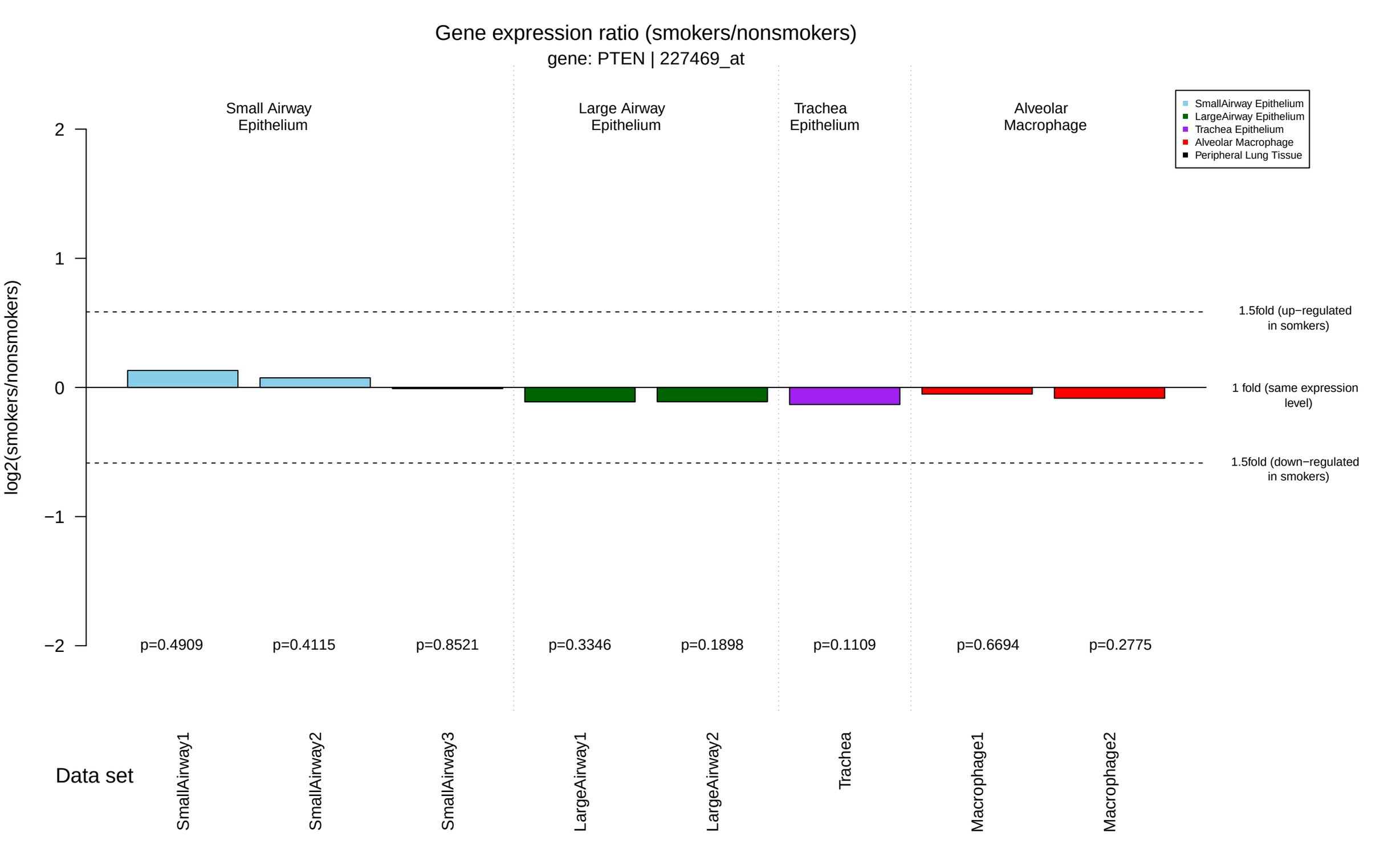
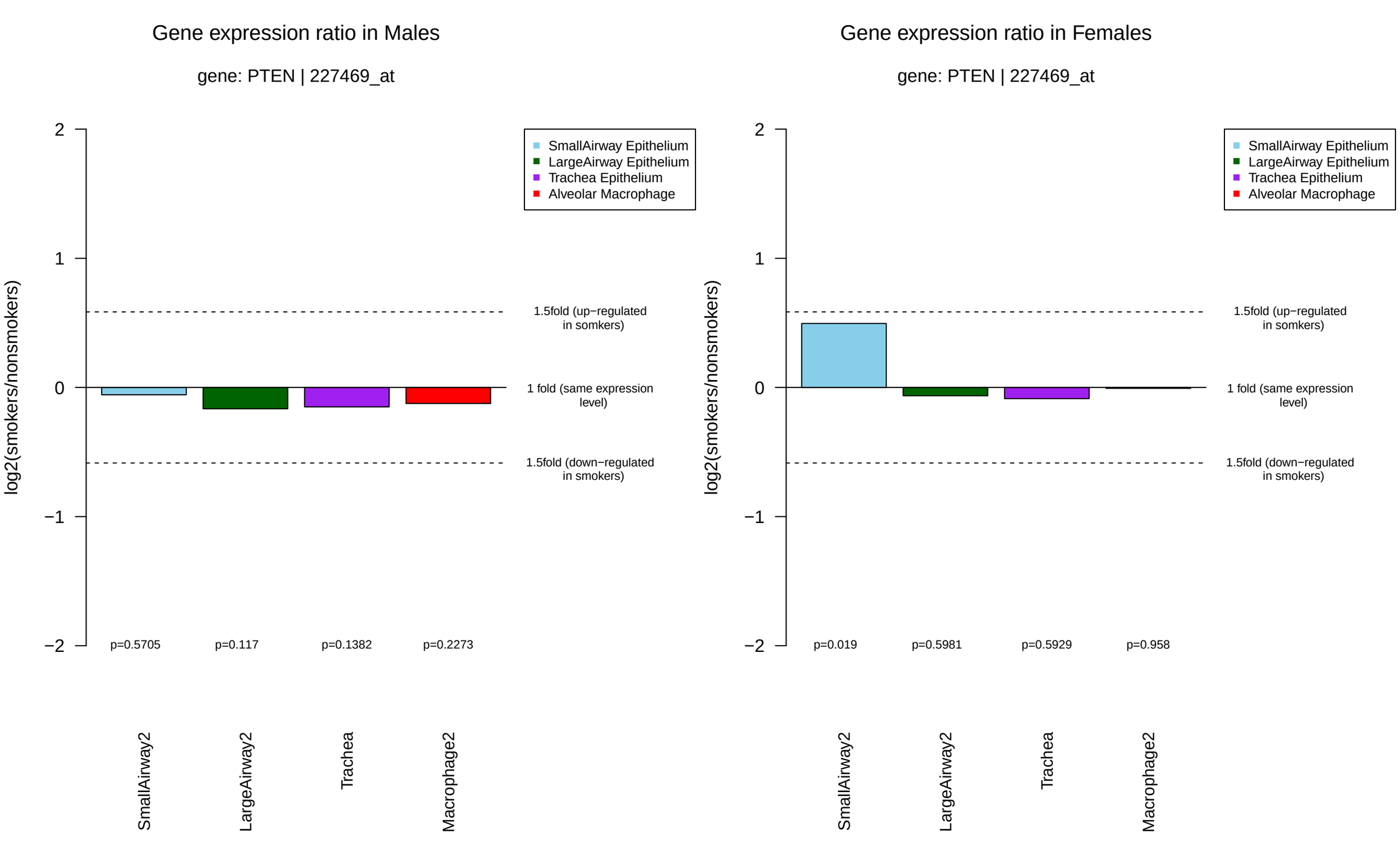
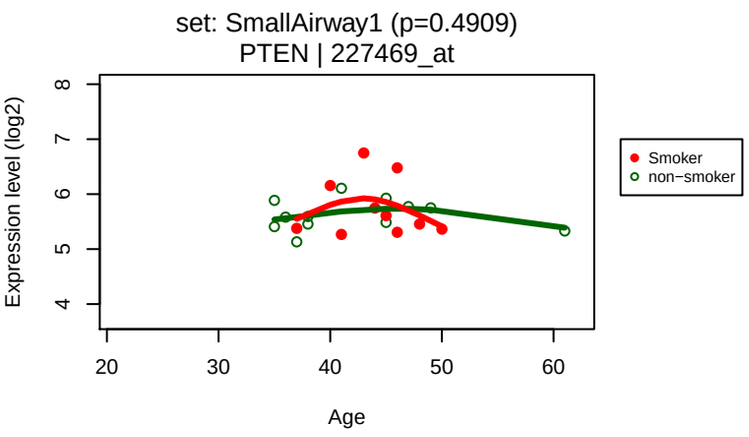
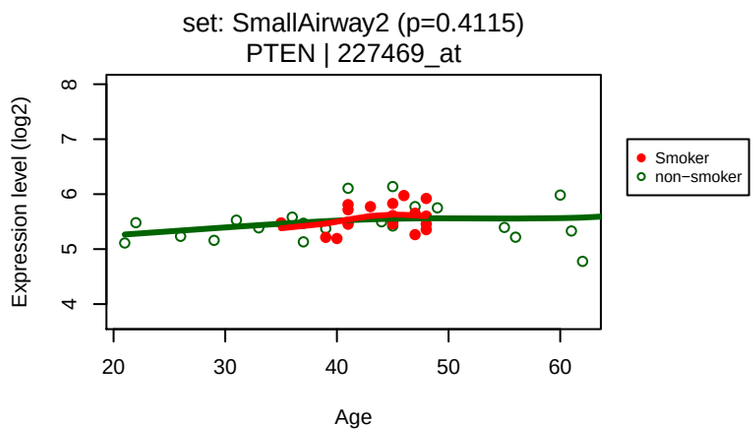
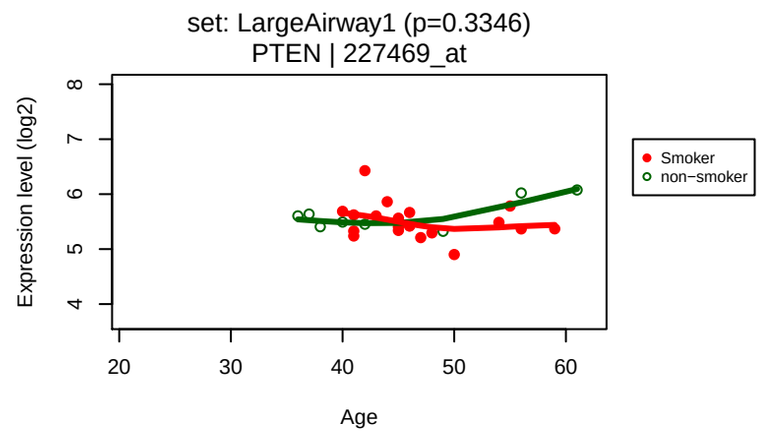

Gene expression of smokers and nonsmokers
gene: PTEN | 227469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4909
p=0.4115
p=0.8521
p=0.3346
p=0.1898
p=0.1109
p=0.6694
p=0.2775
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

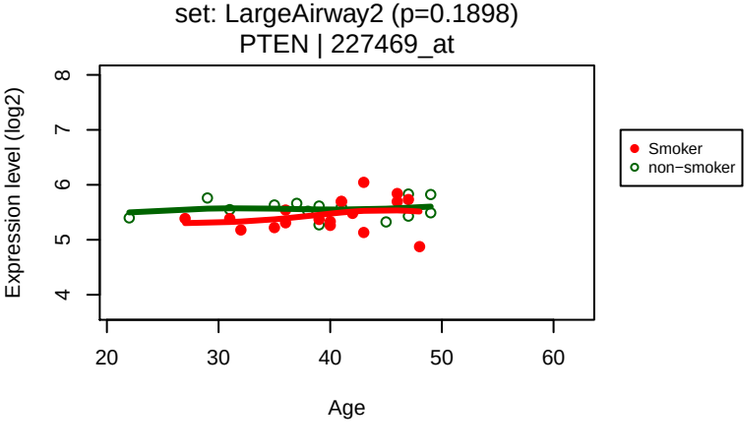

Gene expression ratio (smokers/nonsmokers)
gene: PTEN | 227469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4909
p=0.4115
p=0.8521
p=0.3346
p=0.1898
p=0.1109
p=0.6694
p=0.2775
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

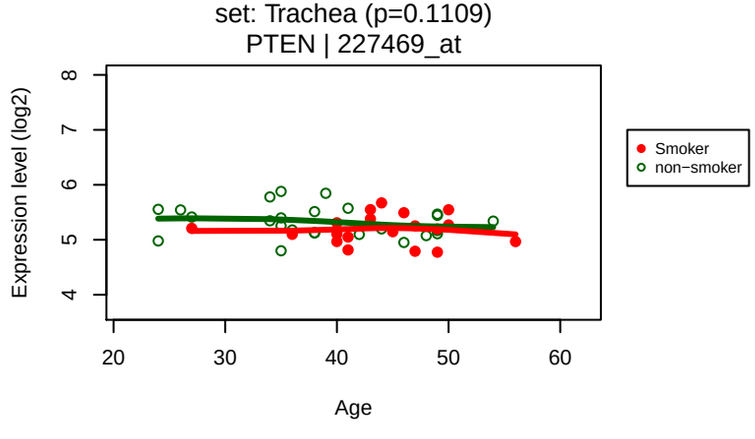

Gene expression ratio in Males
gene: PTEN | 227469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5705
p=0.117
p=0.1382
p=0.2273
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTEN | 227469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.019
p=0.5981
p=0.5929
p=0.958
SmallAirway2
LargeAirway2
Trachea
Macrophage2

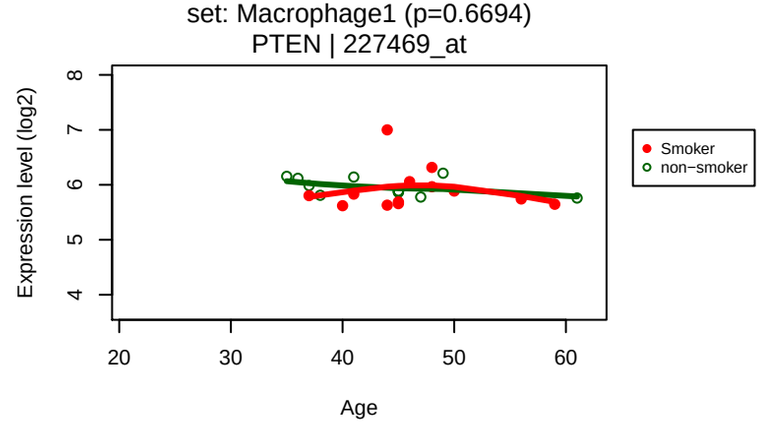

set: SmallAirway1 (p=0.4909)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker

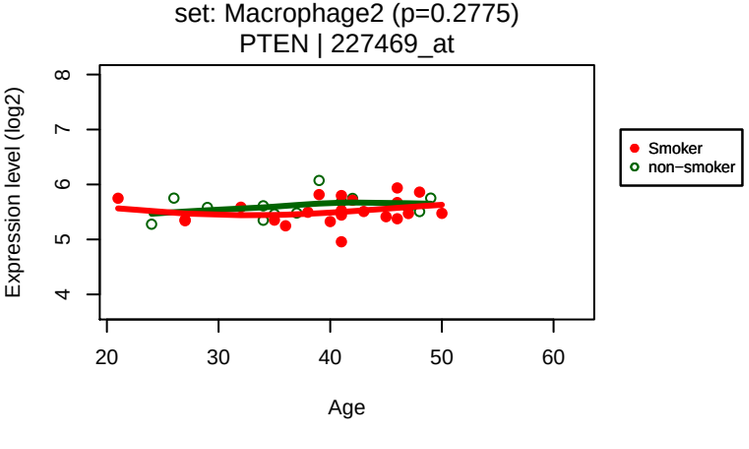

set: SmallAirway2 (p=0.4115)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker

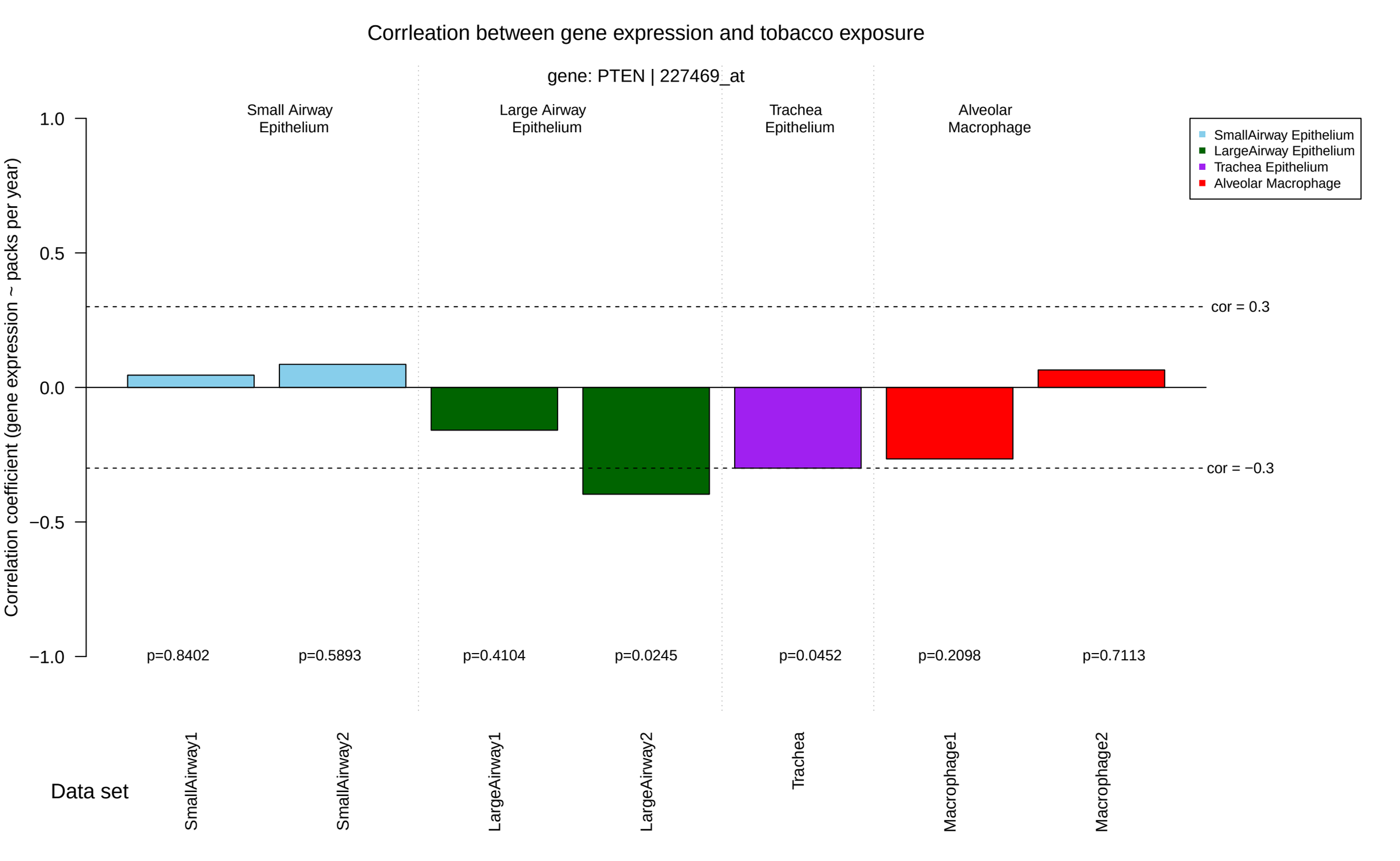

set: LargeAirway1 (p=0.3346)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1898)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1109)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6694)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2775)
PTEN | 227469_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTEN | 227469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8402
p=0.5893
p=0.4104
p=0.0245
p=0.0452
p=0.2098
p=0.7113
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2